

Schizophrenia and Psychotic Disorders

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Session Objectives

- Utilize DSM-V criteria to compare and contrast schizophrenia spectrum and other psychotic disorders.
- Describe the etiology and epidemiology of the schizophrenia spectrum and other psychotic disorders.
- Differentiate between primary psychotic disorders and secondary psychotic disorders (caused by medical problems)
- Describe the evidence-based assessments and treatment approaches for each.



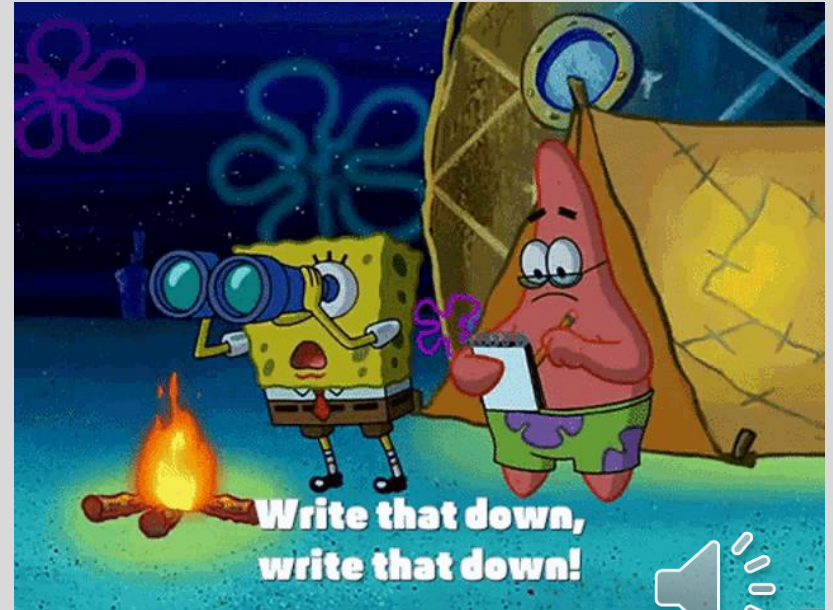
Quick Note:

This lecture will go over high yield topics in Psychiatry that will show up on your board exams. The goal should **NOT** be to simply memorize the DSM-5 criteria and buzz-words, but instead **understand the overall clinical picture**. Very important things you should know will be marked with a **star!**



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Clinical Case I

Gene, a 25 y/o male presents to your emergency department (ED) with chief complaint of being poisoned by his girlfriend and mother. When you enter the room, he has been pacing, but stops upon your entering. You notice that before you can introduce yourself, his lips are softly moving, as if speaking to himself. He steps back when you extend your hand to shake as you introduce yourself. He provides brief, abrupt, short answers after varying latency (sometimes short, sometimes longer). At times he appears distracted, and you need to repeat some questions. He also looks around the room periodically as if distracted or looking for someone/something.



Clinical Case I

When you inquire about past psychiatric hospitalizations, Gene mumbles something under his breath about hating those “shots” and the horrible nurses. When you ask for clarification, he says never mind, and refuses to answer any more questions. He minimizes the concern about being poisoned and asked if he could go smoke marijuana now. He begins to pace again and begins to look more agitated.



Clinical Case I

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- Review of ED chart: No known next of kin and unknown if patient was living with someone or alone. No known past medical history.
- Limited physical exam due to limited cooperation from patient.
- Afebrile. Pulse 90, BP 179/85, RR 20.
- Agitated, little to no eye contact, lungs CTA, CV RRR no murmurs or rubs, abdomen soft.
- Disheveled appearance, poor grooming and hygiene, and little to no eye contact.
- Urine tox positive for THC, no alcohol, no “poison” or other drugs



Clinical Case I

When Gene's mom and girlfriend show up, they note he has been hospitalized a total of 3 times throughout his life and was prescribed olanzapine, but stopped taking it 3 months ago. He has struggled with his mental health starting at 21 years of age. Recently, he has noted hearing voices that "tell him to do bad things".



Question 1

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What are the key points in the case history?



Question 1

What are the key points in the case history?

- Entertaining internal dialogue
- Flat affect
- Appears to be distracted searching for someone/something
- Multiple previous hospitalization with “shots”
- THC positive on urine tox
- Lack of eye-contact
- Hallucinations and delusions: believes his loved ones poisoned him, hearing voices, etc.
- Prescribed an antipsychotic and stopped taking it recently



Psychotic Disorders

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Differential Diagnoses for Non-Psychiatric Causes

- Epilepsy
- h/o head trauma
- Dementias (Alzheimer's, Pick's, or Lewy Body Disease)
- Stroke
- Space-occupying lesions and structural abnormalities of the brain
- Hydrocephalus
- Demyelinating diseases
- Neuropsychiatric disorders
- Autoimmune disorders
- Infections
- Endocrinopathies
- Metabolic disorders
- Chromosomal abnormalities
- Substance use (intoxication or withdrawal)
- Medications
- Toxins



This is all **before** we begin to consider other psychiatric conditions!



Clinical Pearl

Keep in mind, for mental health disorders: you **MUST** rule out other medical conditions first (ex. are symptoms due to depression or are the depressive symptoms due to hypothyroidism?).

In other words, their symptoms are **NOT** better accounted for by another underlying medical condition! (cancer, substance use, etc.)



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Differential Diagnosis for Psychiatric Causes

- Substance induced psychotic disorder
- Bipolar disorder type I, current episode (manic with psychotic features)
- Major depressive disorder with psychotic features
- Schizophreniform
- Brief psychotic disorder
- Schizotypal personality disorder
- Delusional Disorder
- Schizoaffective Disorder



Psychotic Disorders

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Possible Labs You May order for the patient:

- CBC
- Liver functioning tests and ammonia level
- Calcium/phosphorus
- TSH
- ESR—non specific for inflammation if positive
- Antinuclear antibodies—non specific for autoimmune disorders
- Ceruloplasmin or urine copper to rule out Wilson's disease especially if they have physical signs.
- HIV screening—history of IV drug use, other STD
- FTA-Abs for syphilis (RPR not sufficient)—history indicates testing
- Vitamin B and folate
- UA
- Urine Drug Screen
- EEG—suspect delirium or neurologic findings on exam
- Brain imaging especially if have neurologic findings on exam

Massachusetts General Hospital Comprehensive Clinical Psychiatry, Second Edition, Chapter twenty-eight, Tests and Laboratory Findings



Question 2

Preliminary Lab results are unremarkable. Patient begins yelling, threatening to hit anyone who gets close to him and banging on the walls.

What do you want to do now? Has your differential changed?



Question 2

Possible correct answers:

Admission to a psychiatric hospital: possibly initiate a 72 hour hold if he is unwilling to go into the hospital on a voluntary basis

Verbal De-escalation: Use communication strategies to calm the patient and reduce agitation.

Medical Intervention: Administer medications, to sedate the patient. Consider intramuscular (IM) options for moderate to severe agitation. While drug cocktails used to be used, new guidelines recommend an IM injection of Ziprasidone (Geodon).

Restraint and Seclusion: No longer recommended. Can set up a 1:1 if there are available personal.



Schizophrenia



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DSM-V Criteria for Schizophrenia:

≥2 symptoms from this list, and **at least 1** must be a psychotic symptom*. Must be present for **at least 6 months**, and ≥ **1 month** of active symptoms

- Delusions*
- Hallucinations*
- Disorganized speech*
- Grossly disorganized or catatonic behavior
- Negative symptoms (flat affect, avolition, etc.)



Schizophrenia



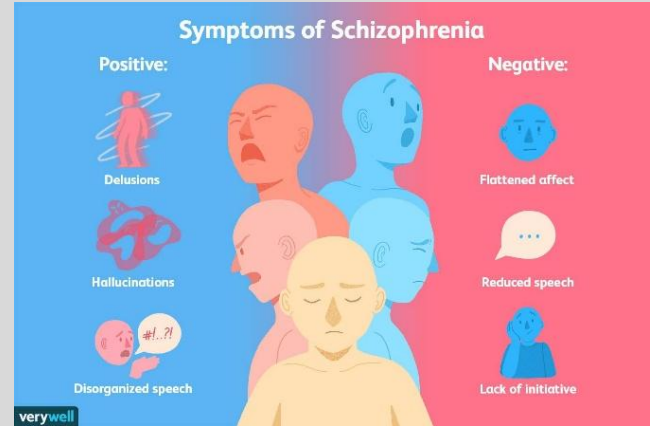
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Positive Symptoms vs Negative Symptoms, what does that mean?

In psychology and psychiatry, “**positive**” means the addition of something that wasn’t there before. It does **not** mean that the symptoms are beneficial at all. In this case, **delusions, hallucinations, and disorganized speech**

Conversely, “**negative**” means taking something away. Thus, negative symptoms include **flattened affect, reduced speech, and lack of initiative/agency**





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Psychotic symptoms include delusions, hallucinations or disorganized speech

- **Delusion:** Fixed, False belief (ex. Family poisoned me)
- **Hallucination:** Perceptions without external stimulus and can occur in any sensory modality (ex. hearing voices)
- **Disorganized speech:** Loosening of associations, derailment, or incoherence



Schizophrenia Spectrum and Other Psychotic Disorders: DSM-5 Selections.

Massachusetts General Hospital Comprehensive Clinical Psychiatry, Second Edition, Chapter twenty-eight, Clinical Features and Diagnosis.



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A quick note on **hallucinations**- they can affect ANY sensation, but some are more common in schizophrenia than others:

- **Auditory:** the most common in schizophrenia
- **Visual:** the second most common form in schizophrenia
- **Olfactory** is a common symptom in temporal lobe epilepsy, uncommon in schizophrenia
- **Tactile:** typical of cocaine or amphetamine use, alcohol withdrawal (ex. Feeling like bugs are crawling on skin)
- **Somatic:** involve a sensation from within the body and are obvious if bizarre. Ekbom's syndrome/delusional parasitosis is an example
- **Gustatory:** rare in schizophrenia





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A quick note on **delusions**: they are false, non-culturally sanctioned beliefs that are held with conviction, even in the face of overwhelming evidence to the contrary

- **Persecutory or paranoid**: most common in schizophrenia or delusional disorder. Belief person is being pursued, stalked, framed, or tricked. (cyberstalking/doxing example)
- **Ideas of reference**: thinks the news or paper are talking about or to them
- **Grandiose**: belief that some agency has taken control of their thoughts, feelings and/or behaviors.
- **Somatic**: part of the body is experienced as being diseased or malfunctioning
- **Nihilism**: exaggerated belief in the futility of everything.
- **Delusions of love**: aka erotomania, patients feel loved by another person, often of higher status, who is merely an innocent bystander
- **Jealousy**: suspecting someone of being unfaithful
- **Delusions of doubles** (aka **Capgras syndrome**): the patient believes that a family member or close friend has been an identically appearing double. More common in dementias.





Schizophrenia

Disorganized behavior can be described as odd, bizarre behavior such as smiling or laughing inappropriately or as if responding to internal stimuli

- Purposeless, ambivalent behavior or movements, and random, intermittent agitation
- Disheveled attire, unusual or inappropriate attire
- Disjointed speech patterns and behaviors reflect disorganized thinking: Perseveration, tangential speech, word-salad, use of neologisms





Schizophrenia

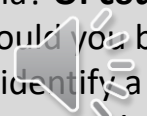
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Catatonia is characterized by the presence of ≥ 3 of 12 symptoms of marked motor disturbance including:

1. **Stupor**- not actively relating to the environment
2. **Catalepsy**-a sudden, temporary loss of muscle control, often triggered by strong emotions like laughter, surprise, or anger
3. **Waxy flexibility**-where a person's limbs remain in whatever position they are placed, after being moved by someone else (like posturing a doll or action figure)
4. **Mutism**- no or very little verbal response (outside of aphasia)
5. **Negativism**-opposition or no response to instructions
6. **Posturing**-spontaneous and active maintenance of a posture against gravity
7. **Mannerism**-odd, circumstantial caricature of normal actions like peculiar ways of dressing or performing everyday tasks,
8. **Stereotypy**-repetitive, abnormally frequent, non-goal-directed movements
9. **Agitation**- specifically NOT influenced by external stimuli
10. **Grimacing**- excessive frowning or teeth bearing
11. **Echolalia**-mimicking another's speech
12. **Echopraxia**-mimicking another's movements

Do you need to memorize each and every aspect of catatonia? **Of course not.** Should you be able to identify a patient presenting with catatonia? Yes.





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Negative Symptoms of Schizophrenia:

- **Blunted or flat Affect**
- **Alogia:** poverty of speech
- **Asociality:** social withdrawal
- **Anhedonia:** lack of interest in things one used to enjoy doing
- **Ambivalence:** inability or extreme difficulty to make decisions
- **Avolition:** lack of interest or engagement in goal-directed behavior

Note: Negative symptoms are easily confused with depressive symptoms, side effects of antipsychotics, or cognitive symptoms. However, they can be just as troubling and disabling to a patient as positive symptoms!





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Treatment: involves use of **antipsychotics**.

- There are **two** main classes of antipsychotics: Typical or **first generation antipsychotics (FGA)** and atypical or **second generation antipsychotics(SGA)**.
- SGA's are commonly used now due to **lower risk** of severe side effects: **Neuroleptic Malignant Syndrome (NMS)** and **Tardive Dyskinesia**.
- **NMS**— defined by elevated temperature, muscle rigidity, irregular pulse, tachycardia, tachypnea, altered mental status and autonomic nervous system dysfunction. Can be lethal if antipsychotic is not stopped and supportive care not given.
- **Tardive Dyskinesia**- chronic neurological syndrome caused by antipsychotic use. Defined by repetitive involuntary movements including grimacing, eye blinking, and other involuntary movement.



Question 3

True or False: Onset of Schizophrenia in childhood traditionally has worse outcomes





Question 3

True or False: Onset of Schizophrenia in childhood traditional has worse outcomes

True!

Typical onset of Schizophrenia is early to mid twenties for males and by late twenties for females and affects about 1 % of the global population

Factors that indicate worse outcomes: Poor pre-morbid functioning, childhood onset, family history of schizophrenia, abnormal brain imaging, insidious onset, and concomitant drug abuse





Delusional Disorder

Delusional Disorder: defined by **1 or more** delusions lasting $\geq$ **1 month(s)** and no other prominent psychotic symptoms (no hallucinations or disorganized speech)

Note: Hallucinations, if present, are mild and directly related to the delusion (e.g., smelling poison if convinced of being poisoned).

- The core feature is the presence of a well-systematized, encapsulated delusional belief
- Has an Insidious onset, frequently occurs in the elderly, and affects women more than men
- Frequently difficult to treat





Schizophreniform and Brief Psychotic Disorder

Similar criteria for symptoms as Schizophrenia, with the major difference being their timing!

Brief psychotic disorder: 1 positive symptom(s) lasting between 1 day and 1 month, usually stress-related. No negative symptoms present

Schizophreniform disorder: 2 symptoms lasting 1–6 months. Impaired functioning not required for diagnosis

Schizophrenia: Symptom onset ≥ 6 months prior to diagnosis; requires ≥ 1 month of active symptoms over the past 6 months.

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Schizophreniform and Brief Psychotic Disorder

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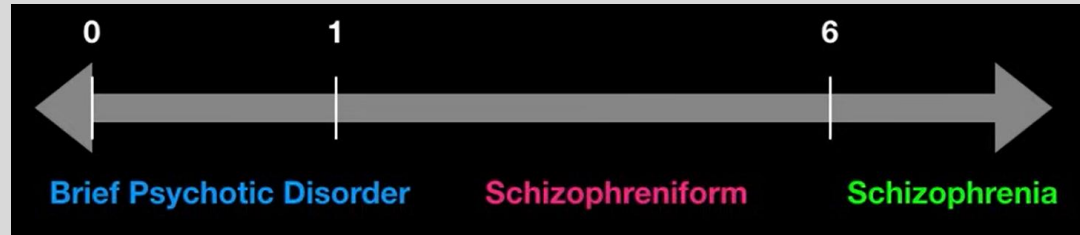
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Think of these three as being on a **continuum of duration and impact:**

Brief Psychotic = sudden onset, short-lived break with reality, full recovery, antipsychotics usually curative.

Schizophreniform = intermediate stage, uncertain prognosis,, also treated with antipsychotics, ~ 1/3 fully recover; most are later diagnosed with Schizophrenia

Schizophrenia = chronic, functionally impairing illness.



Question 4

What is a key difference(s) between Brief Psychotic Disorder and Schizophrenia?

- A. Brief psychotic disorder only involves delusions and not hallucinations
- B. Time frame/duration
- C. Schizophrenia includes positive and negative symptoms, whereas brief psychotic disorder only includes positive symptoms
- D. All of the Above
- E. B and C only



Question 4

What is a key difference(s) between Brief Psychotic Disorder and Schizophrenia?

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- D. All of the Above
- E. B and C only



Schizophrenia

Risk Factors:

Genetic risk

- Family history of schizophrenia in first- or second-degree relative
- Paternal age
- Velo-cardio-facial syndrome (22q11 deletion, different from DiGeorge)

Environmental causes have the best evidence.

- Immigrant populations have the highest risk for developing schizophrenia.
- Insults during pregnancy such as intrauterine infections, particularly influenza
- Maternal starvation



Schizophrenia

Genetic Basis-Two Hit Hypothesis:

- Genes convey a vulnerability to the disease, but **doesn't cause the disease without environmental trigger**
- Most genes don't convey risk of specifically developing schizophrenia, but **confer risk for multiple psychiatric illnesses.**



Clinical Pearl

Is there *another* psychiatric condition?

Clinical Features Suggesting a Secondary Psychotic Illness

- Symptoms of delirium
- Onset of first psychotic episode after age 40
- Rapid development of symptoms in a person without premorbid impairment
- Affect that is normal range and not blunted or flat
Predominance of hallucinations in modalities other than auditory
- Presence of a general medical condition with pathophysiology capable of causing or altering brain functioning
- Evidence of exposure to psychoactive substance capable of causing psychosis



Other Psychotic Disorders

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Schizoaffective Disorders

- Meet criteria for a **mood disorder** (either **unipolar** or **bipolar** type) **AND** criteria for **schizophrenia**.
- To differentiate from a mood disorder with psychotic features, patient must have ≥ 2 weeks of psychotic symptoms without a manic or depressive episode.
- Can have **Schizoaffective disorder, depressed type** (depression symptoms, SIG-E-CAPs) OR **Schizoaffective disorder, bipolar type** (manic symptoms, DIG-FAST)
- **Treatment:** SGAs for both subtypes, add antidepressants for depressed type, add mood stabilizers for bipolar type, ECT for refractory treatment



Test Taking Tip

- Test makers love to have students differentiate between **schizoaffective disorder**, **MDD with psychotic features**, and **Bipolar disorder with psychotic features**
- When answering questions the first part of the name is the primary **psychiatric disorder** while the second part of the name is the **secondary disorder**
- How do I know what is primary or secondary? Look at the question stem carefully to see which symptoms **occur by themselves for AT LEAST two weeks.**
- Ex. Patient presents with **psychotic symptoms** for at least two weeks and has **no** depressive symptoms. Depressive symptoms come and go. **Primary** disorder: schizoaffective, **secondary** disorder, depressed type

Schizoaffective Disorder

Schizo + Affective

Psychotic symptoms

Mood symptoms
(Depressed vs. Bipolar)

Schizoaffective Disorder, depressed type

Major Depressive Disorder, with psychotic features

Schizoaffective Disorder, bipolar type

Bipolar Disorder, with psychotic features





Other Psychotic Disorders

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Postpartum psychosis:

- Very rare, 0.1–0.2% incidence rate. Characterized by mood-congruent delusions, hallucinations, and thoughts of harming the baby or self.
- **Risk factors include:** first pregnancy, family history, bipolar disorder, psychotic disorder, recent medication change.
- **Treatment:** hospitalization and initiation of atypical antipsychotic; ECT for refractory cases.



Schizoid and Schizotypal

These are Type A personality disorders, NOT psychotic disorder (will go more in depth in a separate lecture later)

Schizoid Personality Disorder: Characterized by a profound detachment from social relationships and a restricted range of emotional expression. *Prefers* social withdrawal and solitary activities (vs avoidant), lack of interest in intimacy, limited emotional expression, indifferent to others' opinions (aloof).

Schizotypal Personality Disorder: On the schizophrenia spectrum. May include brief psychotic episodes (eg, delusions) that are less frequent and severe than in schizophrenia. Eccentric appearance, odd beliefs or magical thinking, interpersonal awkwardness. *May* like intimacy, but often find it difficult due to their symptoms and quirks.



Schizoid and Schizotypal

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For schizotypal, I think of Professor Trelawney from Harry Potter or Dr. Brown from Back to the Future

For schizoid, I like to think of Dr. House from the show House MD or Sheldon Cooper from The Big Bang Theory

Schizoid vs. Schizotypal

Schizoid

- NO interest in relationships
- Seclusive to self
- “Loner”
- Works night time jobs

Schizotypal

- Eccentric, magical thinking
- Odd beliefs
- Aloof, dresses weirdly
- Malignant



Clinical Pearls

- Always look at the **teeth**. Poor dentition is quite common in **chronic schizophrenia**.
- Schizophrenia almost always leads to **nicotine** use
- **Hallucinations** in the **absence** of **negative symptoms** and especially **without** the other **positive symptoms** is almost always going to be due to a **substance or a medical cause**.
- Substance use is quite common in schizophrenia, often times used as a **coping skill** or way of managing the illness and not a cause. It can be difficult to differentiate.
- Trust your gut and hairs on the back of your neck. **ALWAYS** leave the interview room if you feel threatened or suddenly scared.



Summary Slide

- **Schizophrenia:** A chronic psychotic disorder lasting ≥ 6 months, with both positive (hallucinations, delusions, disorganized speech/behavior) and negative symptoms, causing significant functional impairment.
- **Brief Psychotic Disorder:** Sudden onset of positive psychotic symptoms lasting ≥ 1 day but < 1 month, followed by full return to baseline functioning.
- **Schizophreniform Disorder:** Psychotic symptoms similar to schizophrenia, lasting ≥ 1 month but < 6 months; functional decline is not required, and prognosis varies.
- **Schizoaffective Disorder – Depressed Type:** Features of schizophrenia (psychosis) along with major depressive episodes, with at least 2 weeks of psychotic symptoms occurring without mood symptoms.
- **Schizoaffective Disorder – Bipolar Type:** Schizophrenia features plus manic or mixed episodes (sometimes also depression), again requiring at least 2 weeks of psychosis without mood symptoms.



Summary Slide

- Postpartum Psychosis:** A rare, severe psychiatric emergency occurring within weeks after childbirth, marked by delusions, hallucinations, mood instability, and high risk of harm to self or infant.
- Delusional Disorder:** Presence of one or more fixed, non-bizarre delusions lasting ≥ 1 month, without other prominent psychotic symptoms or major functional impairment outside the delusional context.
- Schizoid Personality Disorder:** Lifelong pattern of detachment from social relationships and limited emotional expression, with preference for solitary activities and little interest in close connections.
- Schizotypal Personality Disorder:** Pervasive social and interpersonal deficits with cognitive/perceptual distortions (odd beliefs, magical thinking) and eccentric behavior, but without full psychosis.



References

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